

WADII BOULILA, Ph.D., HDR, SFHEA

Professor of Computer Science | Director, RIOTU Laboratory | AI Instructor & Executive Educator
Applied AI & Machine Learning · Generative AI, LLMs & Agentic AI · Computer Vision · Autonomous Systems & Robotics · IoT & Edge Intelligence · Trustworthy AI

✉ wboulila@psu.edu.sa | 📞 +966 56 802 1005 | [LinkedIn](#) | wadiiboulila.com | [Google Scholar](#) | [RIOTU Lab](#)

PROFESSIONAL PROFILE

Professor of Computer Science and Director of the RIOTU Laboratory at Prince Sultan University, with more than 20 years of combined university teaching, hands-on technical training, and applied research in artificial intelligence and advanced computing. I have delivered over 200 professional training sessions of 30 hours each to banking, government, and corporate cohorts across Saudi Arabia and Tunisia, alongside a full academic teaching portfolio spanning machine learning, deep learning, big data analytics, cryptography, and software engineering.

My practice is built on three assets that are difficult to combine. First, certified practitioner experience (Microsoft MCSA/MCTS, Azure, IBM Big Data Engineer) earned over six years as a senior consultant delivering national-scale data platforms for government and banking clients. Second, formally recognised teaching expertise — Senior Fellow of the UK Higher Education Academy, Outstanding Teacher Award at PSU (2025), and co-architect of two full AI degree programmes. Third, current research and build authority in precisely the topics organisations now need training on: generative AI, large language models and agentic systems, computer vision, autonomous unmanned systems, edge intelligence, and trustworthy AI — evidenced by six books with Springer and Elsevier, 359 publications, a granted Saudi patent, two registered AI software copyrights, and recognition among the World's Top 2% Scientists (Stanford/Elsevier) for four consecutive years.

As Director of RIOTU I lead a 25-person laboratory that designs, builds, and field-tests real systems: multi-sensor smart palm farms, an autonomous campus shuttle, GNSS-denied and swarm UAV platforms, quadruped and ground robots for inspection and logistics, and speech-driven agentic intelligence that turns spoken Arabic or English mission objectives into verified autonomous robot action at the edge. That build experience feeds directly back into how I teach — programmes are structured around working systems, live datasets, and the failure modes engineers actually meet, rather than generic vendor material. Delivery is in Arabic, English, or French, on-site or online, for audiences ranging from executive leadership to engineering teams.

CREDENTIALS AT A GLANCE

200+ professional training sessions, 30h each	6,000+ contact hours of hands-on instruction	20+ years teaching and training	25+ distinct courses designed and taught	2 AI degree programmes co-designed
25+ researchers led at RIOTU Lab	6 books with Springer and Elsevier	359 publications; 152 journal articles	11,320 citations; h-index 55, i10 177	83 journal papers in Q1 (JCR)
1 granted Saudi patent	2 registered AI software copyrights	12 competitive grants from 7 agencies	5 keynotes on agentic AI and action models	Top 2% scientist, Stanford / Elsevier

INSTRUCTIONAL DELIVERY CAPABILITY

- **Volume and scale.** 200+ delivered training sessions of 30 hours each, plus two decades of university teaching. Cohorts have ranged from 8-person specialist teams to 60-student lecture and lab groups.
- **Hands-on lab facilitation.** Every professional session has been lab-based: learners work in a live environment while I circulate, debug, unblock, and adapt pace. This includes diagnosing environment and tooling failures in real time, which is where most technical workshops are won or lost.
- **Remote and hybrid delivery.** Regular online instruction via MS Teams, Blackboard Learn, and Moodle, including distributed cohorts across countries, recorded sessions, and virtual lab support.
- **Mixed-ability audiences.** Cohorts spanning database administrators, developers, data scientists, business analysts, and senior decision-makers.
- **Assessment and certification design.** Learning outcomes, rubrics, and ABET-aligned assessment built for two full AI degree programmes; IT badges and certification pathways embedded into professional tracks so outcomes are measurable and portable.
- **Train-the-trainer and faculty development.** Sessions delivered to faculty audiences on technology-enhanced teaching; mentoring of junior researchers and engineers at RIOTU into independent instructional roles.
- **Public speaking.** 15+ international keynotes and distinguished-speaker invitations across Saudi Arabia, the UAE, Bahrain, China, Malaysia, Morocco, and Tunisia, including a national Digital Government Authority workshop.
- **Challenge and hackathon design.** Design and judging of innovation challenges and hackathons; competitive track record as a participant (CVPR 2025 challenge, ROSHN Hackathon AI & Automation).
- **Multilingual.** Full technical delivery in English, Arabic, or French.

TRAINING PORTFOLIO — DELIVERABLE PROGRAMME TRACKS

Generative AI, LLMs & Agentic Systems	Foundation models and their limits; prompting and systematic evaluation; retrieval-augmented generation over enterprise document estates; fine-tuning and parameter-efficient adaptation; small language models for constrained and onboard deployment; stateful agents with memory, tool calling and routing; multi-agent delegation and shared state; orchestration patterns for sequential, branching and event-driven workflows; guardrails, validation, retries and human oversight; evaluation of task completion, tool selection and failure recovery. Grounded in my Springer volumes on Generative AI/LLMs and on Agentic Intelligence.
Applied AI & Machine Learning	Machine learning foundations, feature engineering, model selection and evaluation, deep learning architectures (CNN, RNN, transformers), transfer learning, few-shot and self-supervised learning, weight initialisation and optimisation, model lifecycle and deployment. Levels: foundation through advanced practitioner.
Computer Vision & Geospatial AI	Image classification, object detection and semantic segmentation, satellite and UAV imagery analytics, image restoration, dehazing and enhancement, domain adaptation across aerial and satellite sources, visual inspection and monitoring use cases.
Autonomous Systems & Robotics	Perception, localisation and planning stacks; GNSS-denied and terrain-aided navigation; sensor fusion; ROS2 pipelines and dataset generation; multi-robot and swarm coordination; reinforcement learning for legged locomotion; safety supervision and remote monitoring for field deployment.
IoT, Edge Intelligence & Smart Sensing	IoT reference architectures and microservices; acoustic, olfactory and multi-modal sensing; on-device and edge inference; real-time monitoring; predictive maintenance; anomaly detection; distributed operational data pipelines.
Trustworthy AI, Privacy & Security	AI risk and governance frameworks; uncertainty modelling and model reliability; privacy-preserving machine learning; federated and split learning; secure multi-party computation; neuro-symbolic verification; intrusion detection for IoT and industrial networks; responsible deployment in regulated environments.
Data Engineering, BI & Analytics	Data warehousing and dimensional modelling, ETL/ELT pipelines, SQL Server (SSIS, SSAS, SSRS), Power BI with DAX and MDX, dashboard and KPI design, reporting governance, query performance and database administration, analytics enablement for non-technical decision-makers. Backed by MCSA certification and national-scale delivery.
Executive & Leadership Briefings	AI strategy and operating models; bridging the AI implementation gap; build-versus-buy and capability roadmaps; AI talent development and organisational readiness; sector-specific opportunity mapping; alignment with Saudi Vision 2030 and national digital transformation agendas.

SIGNATURE PROGRAMME BLUEPRINTS

- **Generative AI and Agentic Systems for the Enterprise — 5 days, technical.** Day 1: foundation models, architectures, and realistic capability boundaries. Day 2: prompting, structured outputs, and evaluation harnesses. Day 3: retrieval-augmented generation over the client's own document estate. Day 4: agentic patterns — state, memory, tool calling, routing, multi-agent delegation, orchestration choices. Day 5: guardrails, derailment prevention, cost and latency management, and a supervised build sprint on a client use case.
- **Applied Machine Learning for Data Teams — 30-hours.** Problem framing and data readiness; supervised and unsupervised methods; deep learning essentials; handling scarce and imbalanced data through transfer, few-shot, and self-supervised learning; evaluation beyond accuracy; deployment, monitoring, and drift. Assessed by a capstone on client data.
- **Data Platform and BI Capability Programme — 30 hours per track, multi-cohort.** Separate tracks for administrators (database administration, security, performance), engineers (data warehouse implementation, SSIS/SSAS pipelines), and analysts (data models and reports, Power BI, DAX, KPI design). Designed for institutions standardising reporting across many sites — the structure used for the 174-centre national rollout.
- **Trustworthy AI and Data Governance for Regulated Sectors — 3 days, mixed audience.** AI risk taxonomies and model assurance; uncertainty and reliability; privacy-preserving and federated approaches for data that cannot leave the institution; security of AI-enabled systems; governance operating model, roles, and review gates; documentation and audit readiness.
- **AI for Leadership — half-day executive briefing.** Where value has actually been realised versus where it has not; the implementation gap and its organisational causes; capability, talent, and sourcing decisions; a structured portfolio view of candidate use cases with effort and risk mapped against return.

AGENTIC AI — SYSTEMS, RESEARCH, SUPERVISION & DISSEMINATION

Agentic AI is the centre of my current programme at RIOTU: deployed systems under evaluation, a Springer monograph, supervised graduate research, invited keynotes, and the professional teaching track built directly from them.

Flagship system — Edge Multilingual Agentic AI for Safe Autonomous Robotic Inspection and Exploration

Project Lead and System Architect, RIOTU Laboratory, 2026–Present

Designed and implemented a multilingual, speech-driven agentic platform that lets a user assign high-level exploration, navigation, and inspection missions to a **Unitree Go2 quadruped robot in Arabic and English**. The platform converts natural-language objectives into safe, executable robotic missions through speech recognition, stateful planning, mission memory, retrieval-augmented generation, structured tool calling, and adaptive execution.

- **Agentic mission layer.** Decomposes complex objectives into ordered subtasks and invokes validated **ROS 2** and **Unitree SDK** tools for navigation, exploration, visual inspection, evidence collection, anomaly reporting, and return-to-base operations.

- **Grounded planning.** Environmental maps and inspection checklists are integrated into the planning loop, so missions are grounded in the actual site and visual observations are interpreted against mission-specific criteria rather than generic prompts.
- **Adaptive execution and safety.** Mission-state tracking, obstacle-driven replanning, tool retries, execution monitoring, confidence-based escalation to a human operator, and generation of both spoken and structured inspection reports.
- **Edge deployment.** Deployed on **NVIDIA Jetson Orin NX**, with perception, speech, and small-language-model components optimised to run within the platform’s compute, memory, and power budget.
- **Evaluation.** Assessed on inspection missions against mission completion, correct tool selection, navigation success, failure recovery, and safety-policy compliance.

Trustworthy onboard agentic cyber defence for UAV swarms *Supervised graduate research, RIOTU Laboratory*

- **Small language models** performing onboard reasoning under strict compute and power budgets, so defensive decisions are made in flight rather than offloaded to a ground station that may be unreachable or compromised.
- **Neuro-symbolic verification** checking each agent decision against formal safety and mission constraints *before* execution — a hard gate on agent action rather than a post-hoc review.
- **Byzantine-resilient coordination** keeping the swarm operating correctly when individual agents are compromised, degraded, or behaving maliciously.

Agentic capabilities demonstrated across this work

Stateful planning, memory and control flow	Mission memory maintained across multi-step inspection runs; progress monitoring and contextual carry-over between subtasks; dialogue-level clarification when objectives are underspecified.
Tool calling and validated action interfaces	Structured invocation of ROS 2 and Unitree SDK tools for navigation, exploration, inspection, evidence capture, and return-to-base — with validation at the tool boundary.
Retrieval and external grounding	Environmental maps and mission-specific knowledge retrieved to ground plans and to evaluate observations against expected state.
Task decomposition and routing	Ambiguity resolution, decomposition of high-level objectives into ordered subtasks, and selection of the appropriate tool and action path per subtask.
Multi-agent coordination and shared state	Byzantine-resilient swarm coordination; distributed task allocation and resilient communication across cooperating aerial platforms under degraded links and partial failure.
Guardrails, validation and human oversight	Neuro-symbolic verification against formal safety and mission constraints before execution; confidence-gated escalation to human confirmation; safety-policy compliance as a measured outcome.
Failure recovery	Replanning around obstacles, retry of failed actions, and graceful degradation rather than mission abort.
Evaluation and observability	Explicit metrics for planning correctness, tool-selection accuracy, mission completion, failure recovery, latency, and resource consumption; published empirical studies of LLM capability boundaries and failure modes.
Efficiency and edge deployment	Small language models and optimised perception and speech components running on embedded GPU hardware within fixed power and latency budgets.
Architecture and framework trade-offs	Lead-authored Springer monograph on agentic intelligence plus two volumes on generative AI and LLMs, covering orchestration patterns, agent architectures, and where each approach breaks down.

Authored works

- *The Rise of Agentic Intelligence: From Theory to Real-World Applications* (Springer, 2026) — **lead editor**. Reasoning and planning loops, memory, tool invocation, multi-agent delegation, orchestration frameworks, guardrails, and evaluation, with worked deployable examples.
- *Generative AI and Large Language Models*, Volumes I and II (Springer, 2025–2026) — **co-editor**. Model capabilities, adaptation strategies, orchestration, and enterprise adoption patterns.
- *Edge Intelligence: Advanced Deep Transfer Learning for IoT Security* (Elsevier, 2025) — **co-editor**. On-device inference and security for resource-constrained deployment.

Invited keynotes and talks on agentic AI and large action models

- “The Next Frontier of AI: Beyond Understanding, Toward Action” — 3rd VRISP, China, 2026.
- “Teaching Machines to Act: Large Action Models and the Era of Agentic AI” — 4th SmartTech, Tunisia, 2025.
- “The Data-Efficient Path to Action: Self-Supervised Pretraining for Large Action Models” — 4th OCTA, Tunisia, 2025.
- “Agentic AI: A New Era of Autonomous and Sustainable Intelligence” — Prince Sultan University, Saudi Arabia, 2025.
- “Modern Trends in Artificial Intelligence” — Digital Government Authority workshop, Saudi Arabia, 2026.

Graduate supervision in agentic and language-model systems

- **M.Sc.** — *Trustworthy Onboard Agentic Cyber Defense for UAV Swarms using Small Language Models, Neuro-Symbolic Verification, and Byzantine-Resilient Coordination*.
- **Ph.D.** — *Knowledge-Guided Temporal Multimodal Architectures for Trustworthy Neurological AI* (UTBM France / University of Tunis El Manar, 2026–): multi-step temporal reasoning with multimodal language models.
- **Ph.D.** — *Self-Supervised Domain Adaptation based on Deep Graph Learning* (University of Manouba, 2022–): representation learning under scarce annotation.
- **Engineering team supervision** — the multilingual agentic robotics platform above, spanning speech, planning, tool integration, and edge optimisation.

Community leadership and dissemination

- **Organiser and Chair**, “Agentic Edge Intelligence for the Internet of Things”, IJCNN 2026.
- **Organiser and Chair**, “Distributed Learning and Intelligent Systems: Advancing Privacy and Scalability for IoT and Edge Networks”, IJCNN 2025.
- **Organiser and Chair**, “AI for Autonomous Unmanned Systems” (AAUSS), IWCMC 2023, 2024, 2025.
- Published surveys on LLM–diffusion fusion, temporal reasoning with multimodal LLMs, and federated learning for intelligent transportation, all in *Computer Science Review* (IF 17.5); empirical studies of LLM capability limits in *IEEE Access* and *Heliyon*. Full entries in the publications section below.

APPLIED AI SYSTEMS, ROBOTICS & AGENTIC PLATFORMS

Further systems designed, built, and field-tested under my direction at the RIOTU Laboratory, alongside the agentic platforms described in the preceding section. These are the working references behind the training tracks above.

Smart palm farm — multi-modal early detection of Red Palm Weevil

- End-to-end precision-agriculture platform for large Saudi palm farms, combining **acoustic sensing** (deep learning classification of infestation signatures in the trunk) with **olfactory/gas sensing** (volatile organic compound signatures emitted by infested palms) for earlier and more robust detection than either modality alone.
- Protected by a **granted Saudi patent (SAIP SA 18938 B1, 2024)** and currently under commercial field evaluation, addressing a pest that costs the Gulf date industry hundreds of millions annually.
- Complemented by UAV-based palm tree detection, tracking, and disease mapping using object-level domain adaptation across aerial and satellite imagery, giving farm-scale situational awareness on top of per-tree sensing.
- Funded through the Al Hofuf Company (PI, 100K SAR, 2026–27) and PSU seed programmes for prototyping and field validation; published in *Computers and Electronics in Agriculture* (IF 10.3) and *Ecological Informatics*.

PSU SmartShuttle — autonomous campus mobility

- Cost-effective autonomous shuttle bus operating inside the Prince Sultan University campus: perception and obstacle handling, localisation, route planning, passenger interaction, safety supervision, and remote monitoring of the vehicle fleet.
- Deliberately engineered as a low-cost platform so the architecture transfers to campuses, industrial sites, and closed-community deployments rather than remaining a research demonstrator. Co-PI; PSU Technology Development, 100K SAR, 2026–27.

Unmanned aerial and ground systems

- **GNSS-denied UAV navigation**. Terrain-aided localisation and sensor fusion for flight where satellite positioning is unavailable or spoofed — TERCOM-EKF and vision-based approaches, CNN-LSTM-attention and Kolmogorov–Arnold network estimators, and a two-stage residual extended Kalman filter. Funded by the Prince Sultan Defense Studies and Research Center (300K SAR); published in *Satellite Navigation* (IF 12.4) and *Engineering Applications of Artificial Intelligence* (IF 9.0).
- **Autonomous swarm UAVs**. Multi-agent coordination, distributed task allocation, and resilient communication across cooperating aerial platforms, including behaviour under degraded links and partial failure.
- **Drone delivery and logistics**. Route and battery optimisation, scheduling, and mission planning for delivery and monitoring fleets, including drone-assisted surveillance and infrastructure inspection at scale.
- **Quadruped robotics — inspection and exploration**. QuadroRL: reinforcement learning for robust terrain traversal by search-and-rescue and inspection quadrupeds, targeting unstructured and hazardous environments where wheeled platforms fail.
- **Ground robots for warehouse and service tasks**. Indoor autonomous navigation, mapping, and task execution for logistics handling and service delivery in structured facility environments.
- **Open tooling**. ROS2-based open-source framework for outdoor UAV dataset generation, released to support reproducible field research; UAV-based traffic and parking analytics frameworks (TAU).

Industrial and enterprise AI

- **HydroGuard** — AI-powered predictive maintenance for green hydrogen electrolyzers; registered software copyright (SAIP, 2026). Condition monitoring and remaining-useful-life estimation for high-value industrial assets.
- **Attention-enhanced dual-branch enhancement network** for dynamic light enhancement in night-time imagery; registered software copyright (SAIP, 2025); 2nd place, CVPR 2025 Night Photography Rendering Challenge.
- **Smart water monitoring (SmartWater)** — service-oriented, sensor-cloud architecture for real-time water-environment monitoring; Ministry of Education, 350K SAR.
- **Healthcare and diagnostics** — brain-disorder detection and patient monitoring, federated and privacy-preserving diagnostic pipelines, and early-stage COVID-19 respiratory screening (PI, 200K SAR).

CLIENT & SECTOR DELIVERY EXPERIENCE

- **Government & public sector**: Digital Government Authority, Saudi Arabia (keynote, “Modern Trends in Artificial Intelligence”, 2026); Ministry of Social Affairs; National Health Insurance Fund; National Social Security Fund; National Pension and Social Insurance Fund; National Observatory of Employment and Skills; Research and Social Studies Centre.
- **Banking & financial services**: National Agricultural Bank; Zitouna Bank; Bank of the States of Central Africa — training in BI solution design, SQL Server administration, reporting and analysis services, and application development.
- **Corporate & technology**: Sungard (US); Business & Decision (France); GIZ GmbH (Germany); Integration Objects; Prologic; Arabsoft; Groupe Ozeol; ITAcademy; IMS-CIFOP; Continental; Mediane; BPRO; iPrecision (Czech Republic).
- **Defence and strategic research**: Prince Sultan Defense Studies and Research Center (PSDSARC) — funded UAV navigation programmes and applied autonomy research.

- **Higher education & faculty development:** Prince Sultan University; Taibah University; ISAMM and the Aviation College of Borj El Amri — including faculty-facing sessions on technology-enhanced teaching.
- **National-scale data capability:** Designed and rolled out BI and analytics solutions deployed across 174 training centres of three national agencies (ATFP, ONTT, AVFA) and the social security funds, combined with administrator and end-user training to sustain adoption after handover.

PROFESSIONAL EXPERIENCE

Prince Sultan University — Faculty & Research Leadership

College of Computer and Information Sciences

2021–Present

Riyadh, Saudi Arabia

- **Professor of Computer Science** (2025–Present); previously Associate Professor (2022–2025) and Senior Researcher (2021–2022). Teach Machine Learning and Big Data Analytics at undergraduate level; recipient of the CCIS Outstanding Teacher Award (2025).
- **Director, RIOTU Laboratory** (2024–Present): lead PSU's first and largest multidisciplinary AI and intelligent systems laboratory, with 25+ researchers, engineers, postdoctoral fellows, and graduate students across LLMs and generative AI, computer vision, unmanned aerial and ground systems, robotics, IoT, and edge intelligence.
- Direct the laboratory's research agenda, GPU-backed experimental infrastructure, field-trial programme, recruitment and mentoring, IP capture, partnership development with government and industry, and competitive grant portfolio.
- Run the RIOTU Distinguished Talk Series and the lab's training and outreach programme — a recurring channel for technical workshops serving students, faculty, and external partners.
- Core member of the academic committees establishing PSU's B.Sc. and M.Sc. programmes in Artificial Intelligence (2023–2026): programme design, course sequencing, learning outcomes, and ABET-aligned assessment.
- Member of the CCIS Research and Innovation Committee (2025–2026); Chairman of the CCIS Expo Committee (2023–2024); member of the Quality Committee (2023–2024).

Senior IT Trainer — International Training Centres

Independent and partner-delivered corporate training

2009–2018

Tunisia / Saudi Arabia

- Delivered 200+ training sessions of 30 hours each across business intelligence, databases, web development, and programming for banking, government, NGO, and corporate clients.
- Core curricula: implementing and maintaining SQL Server Reporting, Analysis and Integration Services; designing BI solutions with SQL Server; data warehouse implementation; data models and reports; database administration; querying with Transact-SQL; C# and .NET development; Windows Communication Foundation; introduction to Oracle databases; introduction to cryptography.
- Trained mixed-seniority cohorts from database administrators and developers through business analysts to decision-makers, with emphasis on transfer into the client's own reporting and data environment.

Senior Consultant — Business Intelligence and Data Platforms

Sungard (US); Business & Decision (France); GIZ GmbH (Germany)

2009–2015

Tunisia / France / US

- Designed and rolled out BI and analytics solutions for government, banking, and sectoral organisations, including national-scale deployments and the accompanying training programmes.
- Delivered BI solutions for the Agency for Employment and Self-Employment, the social security funds (CNRPS, CNAM, CNSS), and sectoral bodies; implemented database backup and recovery strategy for an industrial group.
- Designed and developed web portals and digital platforms for governmental and sectoral organisations, combining implementation with capability transfer to client teams.

Associate / Assistant Professor of Computer Science

Taibah University, CCSE; ISAMM, University of Manouba

2012–2021

Madinah, Saudi Arabia / Tunisia

- Taught and coordinated courses in big data analytics, web development, database administration, cryptography, software engineering, and object-oriented design; introduced Machine Learning and Big Data Analytics as new offerings.
- Chaired the Graduation Projects, Information Technology, and Administrative and Technical Affairs committees at Taibah University (2015–2021); coordinated the B.Sc. programme in computer science at ISAMM (2012–2015).

Department Chair & IT Leadership Roles

Aviation College of Borj El Amri (Dept. Chair; Application Development Centre; Regional Maintenance Centre; Network/Systems Engineer)

2005–2012

Tunisia

- Led teaching, systems, and application development functions in an aviation-oriented technical institution; taught data fusion, algorithms and data structures, computer networks, multimedia, and programming in C, C#, and MATLAB.

CURRICULUM DESIGN, TEACHING METHODOLOGY & SUPERVISION

- **Programme architecture:** Core member of the committees establishing the B.Sc. and M.Sc. in Artificial Intelligence at PSU (2023–2026), covering course sequencing, learning outcomes, and ABET-aligned assessment design — directly transferable to designing corporate academies and national capability frameworks.
- **Method:** Computational-thinking-driven instruction using problem-based learning, case studies, stimulated discussion, and active learning; live research datasets and small real-world projects in place of synthetic exercises; IT badges and certification pathways embedded to give measurable, portable outcomes.
- **Courses taught:** Machine Learning; Big Data Analytics (B.Sc. and M.Sc.); Artificial Intelligence; Deep Learning; Research Seminar on Big Data; Advanced Java; Web Development (HTML5/JavaScript/CSS3, ASP.NET, MVC); Advanced Database Administration; Cryptography; Software Engineering; Object-Oriented Design; Data Fusion; Digital Logic and Computer Organisation; Computer Networks; Algorithms and Data Structures; Computer Ethics; C#, C, and MATLAB programming.

Graduate supervision 11 Ph.D. students supervised or co-supervised (7 completed, 4 in progress) across Tunisia, France, the UK, Saudi Arabia, and Italy; 25+ M.Sc.; 50+ B.Sc.; external examiner for 11 doctoral theses internationally, including Università di Bologna (Italy), King Saud University, King Abdulaziz University, and VIT-AP University (India).

- **Current M.Sc. research — agentic autonomy:** *Trustworthy Onboard Agentic Cyber Defense for UAV Swarms using Small Language Models, Neuro-Symbolic Verification, and Byzantine-Resilient Coordination.*
- **Current doctoral research:** knowledge-guided temporal multimodal architectures for trustworthy neurological AI (UTBM France / Tunis El Manar, 2026–); self-supervised domain adaptation with deep graph learning for remote sensing (Manouba, 2022–); microservices and federated learning for IoT intrusion detection and prevention (Carthage, 2021–); anomaly detection, prediction and risk mitigation for husbandry and health systems (Carthage, 2022–).
- **Completed doctorates:** uncertainty in land-cover change models (Manouba, 2017); gene expression modelling (Stirling, UK, 2018); spatiotemporal big remote sensing data (Paris 8 / Manouba, 2020); biomarkers of chronic and complex diseases (Magna Graecia, 2021); segmentation via uncertain constraint networks (Strasbourg / Manouba, 2022); intelligent microservices for IoT data analytics (Manouba, 2023); IoT and deep learning for environmental services (Edinburgh Napier / Manouba, 2023).

BOOKS, KEYNOTES & THOUGHT LEADERSHIP

Books

[1] Boulila, W., Ammar, A., Abdelkader, M., Benjdira, B., et al. (2026). *The Rise of Agentic Intelligence: From Theory to Real-World Applications*. Springer.

[2] Koubaa, A., Ammar, A., Ghouti, L., Boulila, W., Benjdira, B. (2026). *Generative AI and Large Language Models: Opportunities, Challenges, and Applications, Vol. II*. Studies in Computational Intelligence 1214, Springer.

[3] Ahmad, J., Latif, S., Boulila, W., Koubaa, A., et al. (2025). *Edge Intelligence: Advanced Deep Transfer Learning for IoT Security*. Elsevier.

[4] Koubaa, A., Ammar, A., Ghouti, L., Boulila, W., Benjdira, B. (2025). *Generative AI and Large Language Models: Opportunities, Challenges, and Applications, Vol. I*. Studies in Computational Intelligence 1214, Springer.

[5] Boulila, W., Ahmad, J., Koubaa, A., Driss, M., Farah, I.R. (2024). *Decision Making and Security Risk Management for IoT Environments*. Advances in Information Security 106, Springer.

[6] Boulila, W. (2013). *Prédiction des changements en imagerie satellitaire*. Presses Académiques Francophones.

Selected keynotes and invited talks

- “Modern Trends in Artificial Intelligence” — Digital Government Authority workshop, Saudi Arabia, 2026.
- “The Next Frontier of AI: Beyond Understanding, Toward Action” — 3rd VRISP, China, 2026.
- “Bridging the AI Implementation Gap: From Real Needs to Scaled Impact” — 3rd IBI, Tunisia, 2026.
- “The Data-Efficient Path to Action: Self-Supervised Pretraining for Large Action Models” — 4th OCTA, Tunisia, 2025.
- “Teaching Machines to Act: Large Action Models and the Era of Agentic AI” — 4th SmartTech, Tunisia, 2025.
- “Agentic AI: A New Era of Autonomous and Sustainable Intelligence” — Prince Sultan University, Saudi Arabia, 2025.
- “Deep Learning at Scale for Remote Sensing and IoT” — Distinguished Speaker, Jinan University and Jiangsu University of Science and Technology, China, 2025.
- “Minimal Data, Maximum Impact: Optimizing Deep Learning for Real-World Applications” — University of Sharjah Alumni Conference, UAE, 2025.
- “AI-Powered Early Detection System for Red Palm Weevil” — University of Technology Bahrain, 2025.
- “Rethinking Machine Intelligence: The Rise of Few-Shot Learning” — 2nd IPCAINT, Morocco, 2024.
- “Reimagining the Future of Computer Vision: From Object Detection to Generative Models, and Beyond” — 7th IRICT, Malaysia, 2023.
- “Big Remote Sensing Data Analytics and Uncertainty Modeling” — King Abdullah University of Science and Technology (KAUST), Saudi Arabia, 2022.

Conference and editorial leadership General Chair of three international conferences (SmartTech 2022, Saudi Arabia; SmartTech 2024 and IPCAINT 2024, Morocco); organiser and chair of 20+ special sessions at IJCNN, KES, ICCCI, IWCMC, and IRICT (2022–2026), including “Agentic Edge Intelligence for the Internet of Things” (IJCNN 2026) and “AI for Autonomous Unmanned Systems” (IWCMC 2023–2025). Associate Editor, *Frontiers in Communications & Networks* (IoT and Sensor Networks, 2022–present) and *Int. J. Business Data Communications & Networking* (2021–present). Guest Editor for *IEEE J. Biomedical & Health Informatics*, *Int. J. Remote Sensing*, *Remote Sensing*, *Computer Modeling in Engineering & Sciences*, *Computers, Materials & Continua*, and *Discover Internet of Things*. Reviewer for IEEE TGRS, IEEE TCE, *Artificial Intelligence Review*, *Neural Computing & Applications*, CVIU, KAIS, and *Pattern Analysis and Applications*.

RESEARCH PROFILE & PUBLICATION RECORD

Publications	Citations	h-index	i10-index	Q1 (JCR)	Books
359	11,320	55	177	83 / 152	6
Authorship	Sole author	First author	Second author	Last author	Other
	6	35	60	60	150

Figures as of September 2026; impact factors and quartiles as reported in the latest JCR. World’s Top 2% Scientist (Stanford/Elsevier) for four consecutive years, 2022–2025. Full lists: [Google Scholar](#) / [Scopus](#) / [DBLP](#).

Generative AI, LLMs and agentic systems

- [1] Benjdira, B., Ali, A.M., **Boulila, W.**, Koubaa, A. (2026, in press). Fusing LLMs and diffusion models: A comprehensive survey of progress, challenges, and future directions in generative AI. *Computer Science Review*. **IF 17.5, Q1**
- [2] Chaieb, M., Ben Ameer, N., **Boulila, W.** (2026, in press). A comprehensive review of temporal reasoning with multimodal LLMs: Foundations, challenges, and healthcare applications. *Computer Science Review*. **IF 17.5, Q1**
- [3] Koubaa, A., **Boulila, W.**, Ghouti, L., Alzahem, A., Latif, S. (2023). Exploring ChatGPT capabilities and limitations: A survey. *IEEE Access*. **IF 4.2**
- [4] Koubaa, A., Qureshi, B., Ammar, A., Khan, Z., **Boulila, W.**, Ghouti, L. (2023). Humans are still better than ChatGPT: Case of the IEEEExtreme competition. *Heliyon*.
- [5] Alzahem, A., Latif, S., **Boulila, W.**, Koubaa, A. (2023). Unlocking the potential of medical imaging with ChatGPT's intelligent diagnostics. *27th KES International Conference*.

Autonomous systems, UAV navigation and robotics

- [6] Jarraya, I., Alahmed, F., Abdelkader, M., Gabr, K., Kadri, M.B., **Boulila, W.** (2026, in press). CLAK: CNN-LSTM-Attention and Kolmogorov–Arnold Networks for non-visual UAV localization in GNSS-denied environments. *Satellite Navigation*. **IF 12.4, Q1**
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- [11] Benjdira, B., Koubaa, A., Azar, A.T., Khan, Z., Ammar, A., **Boulila, W.** (2022). TAU: A framework for video-based traffic analytics leveraging artificial intelligence and unmanned aerial systems. *Engineering Applications of Artificial Intelligence*. **IF 9.0, Q1**
- [12] Jemmali, M., Bashir, A.K., **Boulila, W.**, Melhim, L.K.B., Jhaveri, R.H., Ahmad, J. (2022). An efficient optimization of battery-drone-based transportation systems for monitoring solar power plants. *IEEE Trans. Intelligent Transportation Systems*.

Precision agriculture and smart sensing

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Computer vision, remote sensing and geospatial AI

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- [18] Ali, A.M., Benjdira, B., **Boulila, W.**, Koubaa, A. (2025). Dehazing-DiffGAN: Sequential fusion of diffusion models and GANs for high-fidelity remote sensing image dehazing. *IEEE Trans. Geoscience and Remote Sensing*. **IF 9.4, Q1**
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IoT, edge intelligence and cybersecurity

- [23] Ben Atitallah, S., Driss, M., **Boulila, W.**, Ben Ghézala, H. (2020). Leveraging deep learning and IoT big data analytics to support smart cities development: Review and future directions. *Computer Science Review*. **IF 17.5, Q1**
- [24] Hajjaji, Y., **Boulila, W.**, Farah, I.R., Romdhani, I., Hussain, A. (2021). Big data and IoT-based applications in smart environments: A systematic review. *Computer Science Review*. **IF 17.5, Q1**
- [25] Babar, M., Qureshi, B., **Boulila, W.**, Koubaa, A. (2026, in press). Federated learning for intelligent transportation systems: Emerging trends, challenges and future directions. *Computer Science Review*. **IF 17.5, Q1**

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Healthcare AI, privacy-preserving and federated learning

[30] Varone, G., **Boulila, W.**, Driss, M., Kumari, S., Khan, M.K., Gadekallu, T.R., Hussain, A. (2024). Finger pinching and imagination classification: A fusion of CNN architectures for IoMT-enabled BCI applications. *Information Fusion*. **IF 17.4, Q1**

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[35] Safaei, M., Sundararajan, E.A., Driss, M., **Boulila, W.**, et al. (2021). A systematic literature review on obesity: Machine learning approaches for prediction. *Computers in Biology and Medicine*. **1,174 citations**

PATENTS & REGISTERED SOFTWARE

[1] **Patent — Saudi Authority for Intellectual Property, SA 18938 B1 (2024).** *Detection of Red Palm Weevil Infestation in Palm Farms*. Granted Saudi patent covering multi-modal early detection of infestation in date palms, combining acoustic signature analysis with deep learning classification and UAV-supported farm-scale survey. Under commercial field evaluation with agricultural partners.

[2] **Software Copyright — SAIP (2026).** *HydroGuard: AI-Powered Predictive Maintenance for Green Hydrogen Electrolyzers*. Registered software implementing condition monitoring, degradation modelling, and remaining-useful-life estimation for electrolyzer stacks in green hydrogen production.

[3] **Software Copyright — SAIP, 25-12-47001624 (2025).** *Attention-Enhanced Dual-Branch Network for Dynamic Light Enhancement in Night-Time Images*. Registered software for low-light image restoration; the underlying method placed 2nd in the CVPR 2025 Night Photography Rendering Challenge.

FUNDED RESEARCH PORTFOLIO

12 competitive grants from 7 funding agencies; 5 as Principal Investigator, 7 as Co-Principal Investigator. Funders include the Saudi Ministry of Education, the Ministry of Culture, the Prince Sultan Defense Studies and Research Center (PSDSARC), King Abdulaziz University, Prince Sultan University, Taibah University, the Al Hofuf Company, and the UK Global Challenges Research Fund.

Role	Selected project	Funder	SAR	Period
Co-PI	Smart Healthcare for Brain Disorders	Ministry of Education	350K	2021–23
Co-PI	SmartWater: Real-Time IoT Water Monitoring	Ministry of Education	350K	2020–22
Co-PI	Terrain-Aided UAV Navigation in GPS-Denied Environments	PSDSARC	300K	2023–24
PI	COVID-19 Respiratory Disorder Detection	Taibah University	200K	2020–21
Co-PI	PSU SmartShuttle: Autonomous Mobility Platform	PSU Technology Development	100K	2026–27
PI	Early Detection of Red Palm Weevil using Olfactory Sensors	Al Hofuf Company	100K	2026–27

HONOURS & AWARDS

International research standing

- **World’s Top 2% Scientist**, Stanford University / Elsevier — single-year impact 2022, 2023, 2024, 2025 (four consecutive years); career-long impact 2025.
- **Foreign Member**, Luxembourg Academy of Sciences and Arts (LASA), Field of Engineering and Technology, 2026.
- **Top Scientist in Computer Science**, [research.com](#), 2025.
- **Premium Residency — Exceptional Talents Category**, Kingdom of Saudi Arabia, 2024.
- **Global Talent Endorsement**, University of Manchester, United Kingdom, 2022.
- **Best Young Researcher in Computer Science in Tunisia**, Beit El Hikma — Tunisian Academy of Sciences, Letters and Arts, 2021.
- **Best Researcher (all fields) and Most Cited Researcher**, University of Manouba, Tunisia, 2021 and 2022.

Teaching and institutional awards

- **Outstanding Teacher Award**, CCIS, Prince Sultan University, 2025.

- **Top 3 Faculty Members for Most Impactful Research Output**, CCIS, Prince Sultan University, 2024 and 2025.
- **Research Excellence Award**, Prince Sultan University, 2023.
- **Highest Impact Factor Pure Research Lab Award**, Prince Sultan University, 2023.
- **Climate Change Award**, Prince Sultan University, 2023.
- **Best Assistant Professor in Tunisia**, 2012.

Competitions and challenges

- **2nd Place**, Night Photography Rendering Challenge, **CVPR 2025** — the top-ranked conference in artificial intelligence.
- **2nd Place**, ROSHN Hackathon — AI & Automation track, 2025, from 1,800 participants.

Publication awards

- **Top-cited paper awards** (2021–2025): *Computer Science Review*; *Computers in Biology and Medicine*; *Int. J. Imaging Systems and Technology* (Wiley); *Software: Practice and Experience*.
- **Best paper awards**: SmartTech 2024 (3rd Int. Conf. on Smart Systems and Emerging Technologies); ICACIn 2021 (2nd Int. Conf. on Advanced Computing and Informatics); ICOICS 2021 (Int. Conf. on Intelligent Computing and Informatics).

Academic distinctions

- **Habilitation (HDR)** and **Ph.D.** both awarded with *Very Honorable, Highest Distinction*.
- **Best Graduation Project Award**, Aviation College of Borj El Amri, Tunis University, 2005.
- **Highest First-Class Honours** for three consecutive years: 2003, 2004, 2005.
- **Baccalaureate in Mathematics**, ranked in the **top 1%** nationally in Tunisia, 2000.

CERTIFICATIONS, PROFESSIONAL STANDING & TECHNICAL STACK

Professional certifications: Microsoft Azure Fundamentals; MCSA (Microsoft Certified Solutions Associate): SQL Server 2012; MCTS (Microsoft Certified Technology Specialist): SQL Server 2008, Implementation and Maintenance; IBM Big Data Engineer — Mastery Award for Educators (2018); IBM Big Data Engineer — Explorer Award for Educators (2018); Big Data Foundations.

Teaching and professional standing: Senior Fellow, Higher Education Academy (SFHEA), United Kingdom; IEEE Senior Member; Member, Association for the Advancement of Artificial Intelligence (AAAI); Member, ACM; Silver Member, KES International; Adjunct Professor, VIT University, India; Founding Head of the Data Science Group, University of Manouba; Senior Researcher, RIADI Laboratory, University of Manouba (2006–present); Senior Research Fellow, ITI Department, University of Rennes 1, France (2015–2020) and CogBID Laboratory, University of Stirling, UK (2015–2019).

AI and systems stack: Python; PyTorch and TensorFlow; Hugging Face; large and small language models; retrieval-augmented generation and vector stores; agent frameworks and orchestration (LangChain/LCEL, LangGraph); guardrails and evaluation harnesses; Docker and FastAPI service endpoints; ROS 2 and Unitree SDK for robotics and UAV pipelines; multilingual speech recognition and spoken interaction; edge and embedded GPU inference on NVIDIA Jetson Orin; multi-GPU training environments; Git-based collaborative development; Linux and Windows.

Data and enterprise stack: C#, VB.NET, Java, C++, T-SQL, PL/SQL, MATLAB; SQL Server, Oracle, MySQL, MS Access; SSIS, SSAS, SSRS, Power BI, PowerPivot, DAX, MDX, data warehousing; Microsoft Azure; Entity Framework, LINQ, ADO.NET; MVC and MVVM; UML modelling; Blackboard, Moodle, and MS Teams for blended delivery.

EDUCATION & LANGUAGES

- **Habilitation (HDR)** — **Qualification to Lead Research in Computer Science**, National College of Computer Science, University of Manouba, Tunisia, 2019. *Very Honorable, Highest Distinction*. Thesis: *From Remote Sensing Data to Knowledge: Modelling Uncertainties, Analysis and Interpretation*.
- **Ph.D. in Computer Science**, University of Rennes 1, France, 2009–2012. *Very Honorable, Highest Distinction*. Thesis: *Spatiotemporal Uncertain Knowledge Discovery for Land Cover Change Prediction in Satellite Imagery*.
- **M.Sc. in Computer Science**, National College of Computer Science, University of Manouba, Tunisia, 2005–2007. Thesis: *A Fusion-based Approach for Modelling Uncertainties in Satellite Imagery*.
- **B.Sc. in Computer Science** (First Class Honours), Aviation College of Borj El Amri, Tunis University, Tunisia, 2000–2005.
- **Baccalaureate in Mathematics**, School of El Haouaria, Tunisia, 2000 — top 1% nationally.
- **Languages:** Arabic (native); English (fluent); French (fluent); Italian (working proficiency). Training delivered in Arabic, English, or French.